

Aircraft Structural Assembly Modification and Aerodynamic Reinforcement Report

Introduction

The aircraft assembly model was progressively refined within the CATIA Assembly Design environment to improve structural rigidity, aerodynamic continuity, load distribution capability, and overall assembly realism. The design modifications primarily focused on the empennage support structure, wing-fuselage mounting region, and reinforcement of the upper fuselage framework.

The modifications were implemented directly within the assembly environment using feature-based modelling techniques, assembly constraints, and integrated support structures to ensure geometric continuity and structural compatibility throughout the aircraft configuration.

Major modifications introduced during the design iteration process included:

- Reinforced upper head plate structure
- Diagonal structural support struts
- Integrated Y-bar support framework
- Extended lower attachment support region
- Aerodynamic head bracket and wing mount reinforcement
- Internal rib reinforcement system
- Wing root fairing and upper aerodynamic transition surfaces

The resulting configuration forms a unified structural and aerodynamic support system intended to improve load transfer efficiency, reduce stress concentration, and enhance aerodynamic performance.

Objectives of the Modification

The primary objectives of the structural and aerodynamic modifications were:

- Increase rigidity of the upper fuselage and empennage support structure
- Improve force distribution between wing, fuselage, and support members
- Reduce localised stress concentration near mounting regions
- Improve aerodynamic continuity around the wing-fuselage interface
- Reduce drag caused by abrupt geometric transitions
- Increase structural realism and mechanical efficiency
- Maintain geometric symmetry and assembly continuity
- Develop a lightweight but structurally efficient aircraft support framework
- Improve long-term durability and resistance to structural deformation

Head Plate Structural Modification

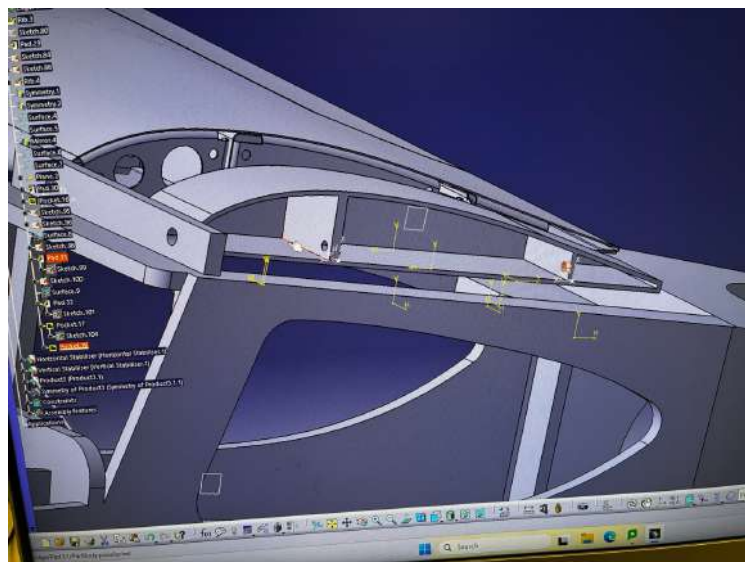
Description

A reinforced upper head plate/interface structure was introduced above the fuselage mounting region to improve structural stability and mounting rigidity.

The component spans laterally across the upper support region and acts as a reinforced interface between the wing support system and fuselage structure.

The plate was integrated directly within the assembly environment and aligned relative to the aircraft centreline.

The plate is made similar to the wing airfoil for aerodynamic efficiency.



Assembly Integration

The head plate was positioned using CATIA assembly constraints including:

- Coincidence constraints
- Contact constraints
- Offset constraints
- Centre-plane alignment constraints

The structure was constrained relative to:

- Upper fuselage geometry
- Wing support mounts
- Internal support ribs
- Upper structural frame components

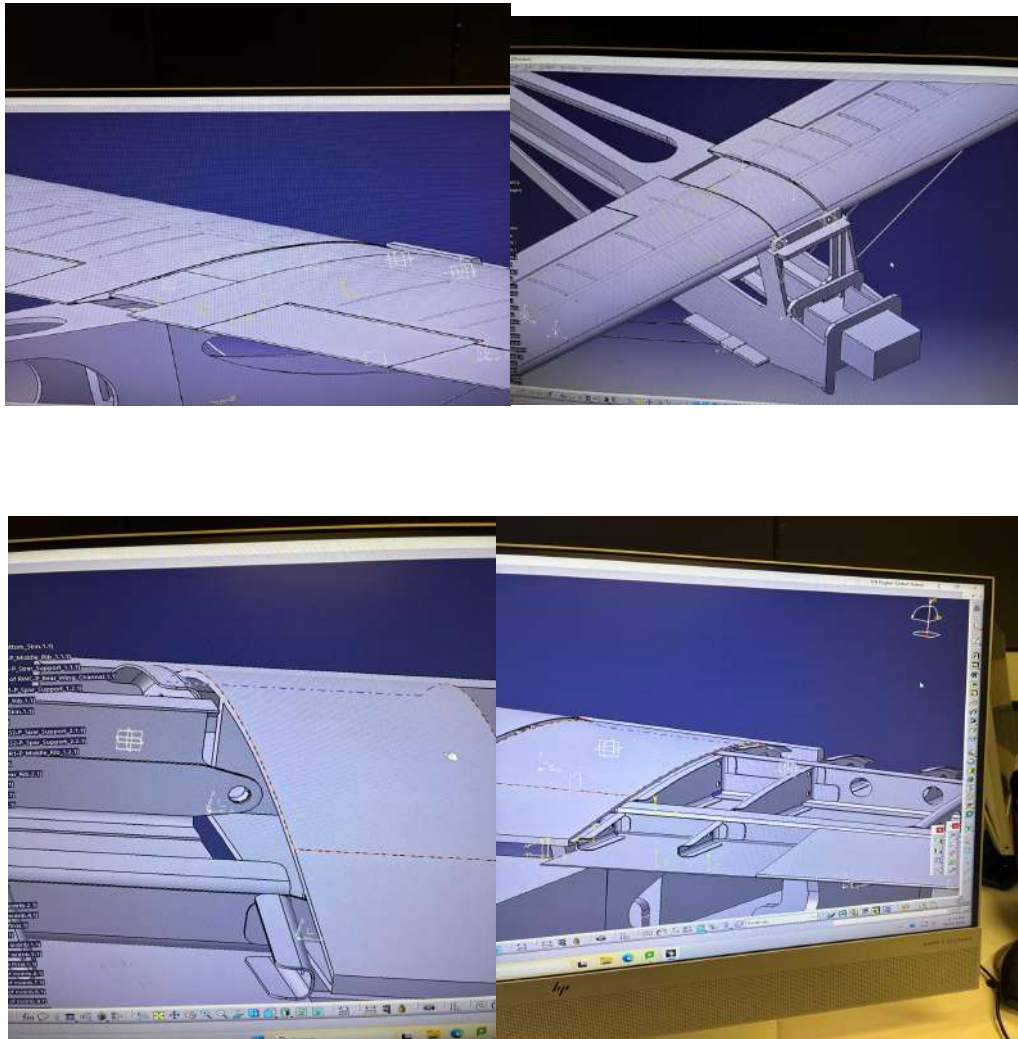
Visual inspection in front, side, top, and isometric views ensured proper symmetry and structural continuity.

Structural Purpose

The reinforced head plate was introduced to:

- Increase rigidity of the wing mounting region
- Improve load transfer into the fuselage structure
- Reinforce the empennage support assembly
- Increase structural continuity across the upper frame
- Improve mounting stability during loading conditions

The modification additionally improved the realism of the aircraft structural arrangement by introducing a practical reinforced mounting interface.



Diagonal Strut Support Assembly

Introduction of Structural Struts

Diagonal support struts were introduced between the lower fuselage mounting region and upper support structure to improve stiffness and structural stability.

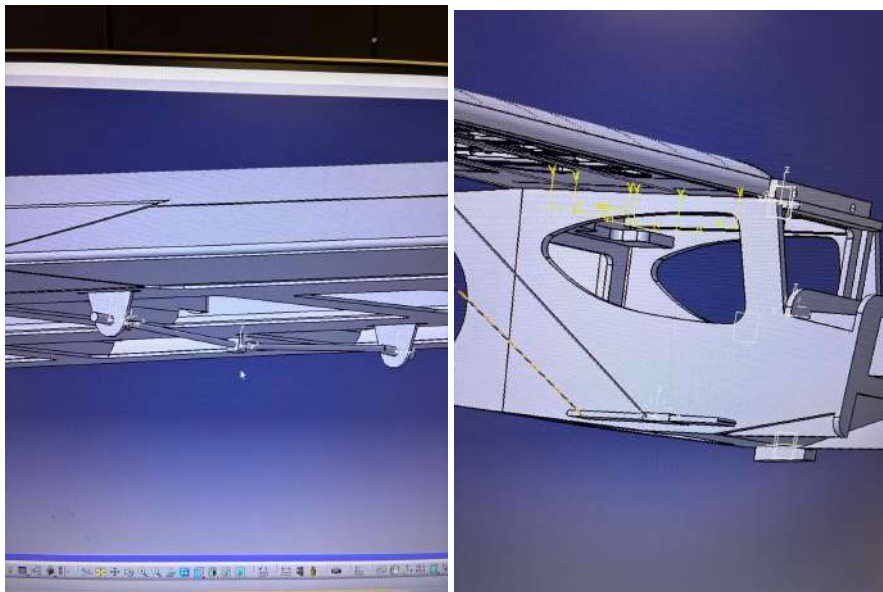
The members were positioned symmetrically relative to the aircraft centreline and constrained directly within the assembly environment.

Structural Configuration

The initial support configuration consisted of inclined support members extending from:

- Lower fuselage support mounts
- Toward the upper empennage and wing support structure

This arrangement formed a triangulated support geometry intended to reduce unsupported span loading and improve rigidity.



Functional Benefits

The diagonal struts improve:

- Bending resistance
- Torsional rigidity

- Load transfer capability
- Structural continuity between upper and lower regions
- Resistance to deformation under loading

The triangulated support arrangement distributes loads through multiple paths and reduces local stress concentration around the mounting interfaces.

Y-Bar Structural Iteration

Assembly-Based Modelling Approach

A further structural refinement was implemented through the development of a custom Y-bar support framework modelled directly within the CATIA assembly environment.

The structure was generated using:

- Circular pad features
- Sequential local axis systems
- Feature-based assembly modelling techniques

This approach enabled accurate alignment relative to surrounding structural components and allowed real-time geometric adjustment throughout the design process.

Y-Bar Geometry

The resulting configuration forms a Y-shaped support arrangement consisting of:

- Two inclined support members
- A central intersecting junction
- A longitudinal connecting support bar
- Additional support members connected near the leading support mount

The geometry creates a semi-trussed support structure beneath the upper fuselage and empennage region.

Structural Purpose

The Y-bar support arrangement improves structural efficiency by:

- Introducing multiple load transfer paths
- Improving stress distribution

- Reducing unsupported member span
- Increasing torsional resistance
- Increasing rigidity beneath the upper support structure

The triangulated arrangement improves the mechanical efficiency of the support system while maintaining lightweight structural characteristics consistent with aircraft structural design philosophy.

Lower Attachment Extension

Description

The lower attachment region of the support assembly was extended downward relative to the initial configuration while maintaining compliance with course design constraints.

Importantly, the fuselage itself was not extended beyond the allowable structural design scope.

Functional Advantages

The extended lower support region provides:

- Improved structural support length
- Better load transfer distance
- Increased support stability
- Potential stepping or boarding functionality
- Potential wheel or landing support interface capability
- Additional mounting possibilities for future design development

This modification further improves structural continuity between the upper support structure and lower fuselage region.

Aerodynamic Head Bracket and Wing Mount Reinforcement

Aerodynamic Head Bracket Design

An aerodynamic head bracket was developed directly within the assembly to support the wing mounting system onto the fuselage structure.

Unlike a conventional flat structural plate, the component was designed with a curved aerodynamic profile to reduce drag and improve airflow continuity around the wing-fuselage interface region.

Wing Support Integration

The aerodynamic head bracket acts as the primary wing mounting interface and transfers wing loading into the upper fuselage support structure.

The arrangement improves:

- Mounting rigidity
- Load distribution capability
- Structural continuity
- Stability around wing mounting regions

The head bracket also integrates directly with the previously developed Y-bar strut support system.

Internal Rib Reinforcement

Internal reinforcing ribs were integrated within the head bracket structure to improve stiffness and structural durability.

The ribs were drilled with 3.5 mm diameter holes to accommodate the wing support mount interfaces.

Hole positioning was defined relative to the rib edges at:

- Vertical edge distance: 8 mm
- Horizontal edge distance: 8.5 mm

These dimensions maintain sufficient material integrity around the mounting region while allowing secure structural fastening.

Structural Function of Reinforced Rib System

The internal rib framework improves:

- Resistance to bending deformation
- Local stiffness near mounting interfaces
- Load distribution across the head bracket

- Resistance to localised stress concentration
- Structural durability during loading conditions

The reinforced arrangement prevents excessive deformation around the wing mounting system.

Wing Root Fairing and Aerodynamic Surface Integration

A further aerodynamic refinement was introduced through the addition of a blended wing root fairing and upper transition surface between the fuselage and wing support structure.

This modification was developed directly within the CATIA modelling environment to improve aerodynamic continuity and surface integration.

Surface Transition Geometry

The fairing structure creates a smooth aerodynamic transition between:

- Upper fuselage structure
- Wing root region
- Reinforced head bracket
- Internal rib support assembly

The curved geometry reduces abrupt geometric transitions and improves airflow continuity across the upper fuselage region.

Structural Integration

The aerodynamic fairing integrates directly with the internal rib support framework and upper structural members.

The support arrangement includes:

- Longitudinal support members
- Internal ribs

- Wing mounting interfaces
- Structural load transfer channels

This integration improves both aerodynamic performance and structural stiffness.

Aerodynamic Benefits

The blended transition geometry reduces:

- Local airflow separation
- Turbulence generation
- Flow disturbance around structural intersections
- Drag caused by abrupt profile changes

The modification improves the aerodynamic efficiency and visual realism of the aircraft assembly.

Integrated Structural Evaluation

Following the integration of all structural and aerodynamic modifications, the assembly demonstrated significant improvement in:

- Structural rigidity
- Geometric stability
- Load transfer efficiency
- Aerodynamic continuity
- Wing root reinforcement
- Resistance to deformation
- Mounting stability
- Structural realism

The combined arrangement of:

- Reinforced head bracket
- Internal rib framework
- Y-bar support structure
- Diagonal struts
- Extended lower support region
- Wing root fairing system

creates a unified lightweight aircraft structural framework with multiple distributed load paths and improved mechanical efficiency.

The integrated system significantly reduces stress concentration and structural wear throughout the fuselage and wing mounting region.

Discussion

The final structural configuration demonstrated improved rigidity, aerodynamic continuity, and force distribution throughout the wing-fuselage support region. The integration of the reinforced head bracket, internal rib structure, diagonal struts, and Y-bar support system created multiple load transfer paths which reduced localised stress concentration and improved structural stability.

To further strengthen the wing mounting interfaces, aerospace-grade fasteners such as AN3 or AN4 series bolts may be used due to their high shear strength, fatigue resistance, and reliability under cyclic loading conditions. For higher load-bearing regions, NAS standard bolts could additionally be implemented to improve tensile performance and structural durability. Locking mechanisms such as Nyloc nuts, castellated nuts, cotter pins, and lock washers would further improve fastening security during vibration and operational loading.

Future work should focus on Finite Element Analysis (FEA) to evaluate deformation, stress distribution, and critical load paths throughout the assembly. Particular attention should be given to the wing root region, rib mounting holes, strut intersections, and lower support attachments where stress concentration may occur.

Further analysis including fatigue and life-cycle assessment could be conducted to evaluate crack initiation, vibration-induced wear, and long-term structural durability under cyclic loading conditions. Margin of safety calculations should additionally be performed for all major support members to validate structural reliability under operational loading.

Aerodynamic refinement through Computational Fluid Dynamics (CFD) analysis could also improve airflow continuity and reduce drag around the wing-fuselage interface and fairing regions. Future iterations may additionally incorporate lightweight composite materials and topology optimisation techniques to further improve strength-to-weight efficiency and overall structural performance.

Conclusion

The aircraft assembly was successfully refined through multiple structural and aerodynamic design iterations within the CATIA Assembly Design environment. The integration of reinforced head brackets, internal rib systems, diagonal support struts, Y-bar load-bearing structures, and aerodynamic wing root fairings significantly improved the rigidity, load transfer capability, and structural continuity of the overall configuration.

The combined support framework created multiple distributed load paths throughout the wing-fuselage interface and upper support structure, reducing localised stress concentration while improving resistance to bending and torsional deformation. The introduction of aerodynamic transition surfaces additionally improved airflow continuity and reduced abrupt geometric changes around critical mounting regions.

The iterative assembly-based modelling approach enabled continuous refinement of structural alignment, mounting integration, and support geometry while maintaining realistic lightweight aircraft design characteristics. The final configuration demonstrates a structurally robust and aerodynamically refined conceptual support system suitable for further engineering development.

Although the current model provides a strong conceptual framework, future analytical validation through Finite Element Analysis (FEA), fatigue assessment, margin of safety calculations, and Computational Fluid Dynamics (CFD) analysis would allow further optimisation of the structure. These future investigations would improve understanding of stress distribution, deformation behaviour, life-cycle durability, and aerodynamic efficiency under operational loading conditions.

Overall, the completed assembly demonstrates an effective integration of structural reinforcement and aerodynamic refinement principles, resulting in a mechanically efficient and visually realistic lightweight aircraft support architecture.

Final Complete Assembly

